

# A Multi-Scale Computational Framework Reveals Evolutionary Constraints Distinguishing Avian Cryptochrome-4 from Related Photoreceptor Proteins

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September 2026

## Abstract

The radical-pair mechanism of cryptochrome-4 (Cry4) is the leading candidate for the avian magnetic compass. Two questions organize the field: whether magnetic sensitivity can be read from a protein's sequence or static structure, and whether the magnetosensor carries a distinctive evolutionary signature. We answer both under pre-registered gates on a 116-species avian panel. The positive result is evolutionary: the avian Cry1 folded domain is canalized at the whole-protein level, with background conservation  $W = 0.9955$  ranking first of six proteins (GAPDH 0.9899, rhodopsin 0.9755, LDHA 0.9718, PKM 0.9675, melanopsin 0.8962; all five pairwise permutation tests significant, four at the  $p = 10^{-4}$  floor), robust under column bootstrap and leave-one-species-out (minimum  $W$  0.9949). A locked cryptochrome-2 paralog control then shows the same extreme canalization ( $W = 0.9941$ , confidence intervals overlapping): canalization is a family-level property of avian cryptochromes, not a magnetosensor-specific adaptation. Every pre-registered probe for a magnetosensor-specific signature falsified - gene-by-phenotype conservation contrast ( $p = 0.29$ ), radical-pair-sphere constraint ( $W_{RP}$  0.9995 vs. 0.9975, intervals overlapping), APC-corrected co-evolution networks (the Cry1 radical-pair-coupled component is absent while Cry2's is present, an inversion), and C-terminal tail charge ( $p = 1.0$  in Cry1) - closing the static sequence/structure/network-descriptor route to a specificity claim. The constraint result is physical: five pre-registered tests of the structural account of magnetic sensitivity, from sequence features to an exactly solvable Haberkorn spin-dynamics pipeline parameterized by predicted structure (validated at 1.2-1.8 Å against AlphaFold DB and the 6PTZ crystal), all failed to separate migratory from sedentary species (LOO AUC 0.21-0.40, within permutation nulls), and the physics-forward back-electron-transfer lifetime model anti-predicted the experimental ordering of Xu et al. (2021) by 19.9×. Pre-registering that failure as the hypothesis, terminal-Trp edge distance as the dominant determinant of magnetic sensitivity is ruled out under the tested model class (Spearman  $\rho = -1.0$ , bootstrap 95% CI  $[-1.0, -0.5]$ ; a 369,495-point sweep recovers the ordering at 0.64% of points, 0/53,703 in the physical subspace). A final locked analysis shows the fourth tryptophan extends the magnetically responsive state ~44-fold with peak sensitivity and field selectivity identical within grid resolution - a free temporal extension that motivates the tetrad's conservation. Together: the avian compass proteins are globally frozen folds, and the migratory advantage must live in degrees of freedom - dynamics, the hyper-fine environment, interaction partners - that static sequence and structure do not capture.

# 1 Introduction

Night-migratory songbirds navigate by a light-dependent magnetic compass whose primary sensor is believed to be cryptochrome-4 (Cry4), a flavoprotein expressed in the retina. The radical-pair mechanism holds that photoexcitation of the FAD cofactor triggers a cascade of electron transfers along a conserved chain of four tryptophans (the Trp tetrad), producing a spin-correlated radical pair whose singlet–triplet interconversion - and hence its chemical yield - is modulated by the geomagnetic field. Xu et al. (2021) demonstrated in vitro that Cry4 from the European robin is magnetically sensitive, and more so than Cry4 from chicken and pigeon, and that the fourth tryptophan prolongs the magnetically sensitive state.

Despite two decades of radical-pair theory, no computational tool maps a cryptochrome sequence to a predicted magnetosensitivity. Building such a tool matters for three reasons: it would make the field’s central hypothesis falsifiable at genomic scale; it would allow ancestral and extinct magnetosensory phenotypes to be reconstructed; and it would convert a qualitative biophysical narrative into quantitative, testable engineering knowledge. This paper takes a deliberately hypothesis-driven path. We first build and benchmark an exactly solvable spin-dynamics pipeline (Sections 2–4). We then ask, under three successively better-grounded pre-registered operationalizations, whether sequence or structure alone separates migratory from sedentary birds across a frozen 116-species panel - and report three rigorous negative results (Section 5). Finally, we redirect to a physics-forward replication design, pre-registered before any computation, that asks whether an independently parameterized model can predict the interspecies and fourth-tryptophan dependence measured by Xu et al. without fitting to the phenotype (Section 6). Section 7 then turns the panel itself into the instrument: pre-registered evolutionary constraint mapping establishes whole-protein canalization of avian cryptochromes, localizes it as a family-level pattern through a locked paralog control, and closes the static-descriptor route to a magnetosensor-specific signature.

## 2 Theory

The spin dynamics of a two-radical pair under hyperfine and Zeeman couplings follow the Haberkorn master equation

$$\frac{d\rho}{dt} = -i[H, \rho] - k_S\{P_S, \rho\} - k_T\{P_T, \rho\}, \quad (1)$$

where  $H$  is the spin Hamiltonian, and  $P_S, P_T$  are the singlet and triplet projectors satisfying

$$P_S + P_T = I, \quad P_S = \frac{1}{4}I - \mathbf{s}_1 \cdot \mathbf{s}_2. \quad (2)$$

The singlet recombination yield from an initial singlet state  $\rho_0 = P_S/4$  has the closed form

$$\Phi_S = k_S \int_0^\infty \text{Tr}[P_S \rho(t)] dt = -k_S P_S \cdot (L^{-1} \rho_0), \quad (3)$$

with  $L$  the Liouvillian of Eq. (1). The anisotropy of the yield over field orientations  $(\theta, \phi)$  is

$$\Gamma = \max_{\theta} \Phi_S(\theta) - \min_{\theta} \Phi_S(\theta), \quad A = \Gamma / \langle \Phi_S \rangle. \quad (4)$$

The hyperfine field on each radical is

$$H_{\text{hf}} = \sum_k \mathbf{s}_1 \cdot \mathbf{A}_k \hat{I}_k, \quad (5)$$

and the Zeeman term at geomagnetic strength  $B = 50 \mu\text{T}$  is

$$H_Z = \gamma \mathbf{B} \cdot (\mathbf{s}_1 + \mathbf{s}_2), \quad \gamma = 28.0 \text{ GHz/T}. \quad (6)$$

## 2.1 Back-electron-transfer lifetime (model M1.2)

The interconversion rate is set by back electron transfer from the terminal tryptophan of the tetrad chain to FAD,

$$\tau = \tau_0 \exp[\beta(r - r_0)], \quad \beta = 1.4^{-1}, \quad (7)$$

anchored to  $\tau = 1.1 \mu\text{s}$  for the Cry4 tetrad and  $0.14 \mu\text{s}$  for a triad (locked 2026-09-25, pre-scoring). The effective singlet and triplet rates used in Eq. (1) are

$$k_S = k_T = \frac{1}{2\tau}. \quad (8)$$

## 2.2 Tryptophan-chain geometry

Chain assignment from the crystal structure 6PTZ: W395  $\rightarrow$  W372  $\rightarrow$  W318  $\rightarrow$  W369, FAD-to-surface. Edge distances  $r_{ij}$  between successive indole centroids satisfy

$$r_{ij} = \|\mathbf{c}_i - \mathbf{c}_j\|, \quad \tau_{\text{chain}} \propto \exp(\beta r_{\text{term-FAD}}). \quad (9)$$

## 2.3 Hyperfine parameterization

Hyperfine tensors for FAD $\bullet^-$  and Trp $\bullet^+$  radicals follow the published DFT set of Hiscock et al. (2016): FAD N5, N10, H6; Trp N1, H4, H7 dominate the anisotropy. The effective isotropic couplings satisfy

$$a_{\text{iso}} = \frac{1}{3} \text{Tr}(A_k), \quad \sigma_{\text{eff}}^2 = \sum_k \frac{2}{3} a_k^2 I_k(I_k + 1), \quad (10)$$

where  $\sigma_{\text{eff}}$  sets the S-T mixing rate in the semiclassical limit

$$\omega_{\text{ST}} \approx \sigma_{\text{eff}} \gamma B / \sqrt{\sigma_{\text{eff}}^2 + (\gamma B)^2}. \quad (11)$$

The directional response is maximized when the lifetime matches the mixing time,

$$\tau_{\text{opt}} \approx \frac{2\pi}{\omega_{\text{ST}}}, \quad (12)$$

which for geomagnetic fields and the Cry4 hyperfine set evaluates to the microsecond regime - precisely the regime the tetrad chain reaches via Eq. (7) and the triad does not. This is the mechanistic content of hypothesis H1.

# 3 Methods

## 3.1 Sequence panel

A frozen panel of 36 migratory and 80 sedentary bird species was assembled from RefSeq annotation (NCBI datasets v2alpha, reference\_only), with migratory status grounded in the AVONET trait database (Tobias et al., figshare 16586228). Cryptochrome-4 orthologs were identified by profile-HMM search (HMMER 3.4, EBI phmmer REST,  $E \leq 1.3 \times 10^{-172}$ , rank-1 validated) against the 121-sequence reference alignment, yielding an 11-species Cry4 subpanel.

## 3.2 Multiple sequence alignment and features

The panel was aligned with FAMSA and cross-checked by EBI MAFFT REST. Feature set v1 comprises 116 per-sequence descriptors: Trp-tetrad presence and spacing, FAD-pocket residues, chain-edge distances mapped from the 6PTZ numbering, and physicochemical profiles.

## 3.3 Structure grounding and cross-validation

The tryptophan chain was assigned from the X-ray structure 6PTZ (Zoltowski et al.; COMPND mislabels the construct “CRYPTOCHROME-1”; sequence is 99.6% identical to *C/Cry4 A0A386QUR4* over the aligned region): W395 → W372 → W318 → W369, FAD-to-surface. Every subpanel sequence was folded by ESMFold (ESM Atlas API, 400-aa cap) and, where available, retrieved from AlphaFold DB (model v6). ESM, AFDB, and crystal coordinates agree to 1.2–1.8 Å over the chain region, establishing that structure-derived distances are pipeline-invariant.

## 3.4 Spin-chemistry scoring

For each sequence, chain-edge distances feed the back-ET lifetime model M1.2 (Eq. (7)); the Haberkorn equation (1) is integrated in the closed form of Eq. (3) over a 65-orientation Lebedev grid to produce an anisotropy amplitude  $A$  (Eq. (4)). The locked decision rule is  $AUC \geq 0.75$  of  $A$  separating migratory from sedentary panel members (pre-registered before scoring).

## 3.5 Structural landscape

To place 6PTZ in context, the cryptochrome fold census was enumerated by RCSB full-text search (34 structures) with per-entry resolution and method from the RCSB data API; FAD binding-site geometry, ligand identity (FAD, residue 501, chain A; 95.4% observed residues), and secondary structure (31 helices, 10 strands) were grounded through the PDBe API. Domain architecture was independently confirmed by InterPro (cryptochrome/photolyase class-1) after correcting a UniProt mis-annotation of the erCry4 reference (Q6LML8  $\neq$  Cry4; correct accession A0A2I4SZI9).

# 4 Results

## 4.1 Benchmark battery (G3)

The solver was validated against five pre-registered benchmarks (results/benchmarks.md): B1 (Ritz  $\theta$ -curve reproduction) PASS; B2 (Hiscock et al. 2016 field-strength response spike) PASS; B5 (isotropic-field zero response) PASS; B3a (monochromatic attenuation 93.8–93.9%) PASS; B3b FAIL (7.3% residual vs. 5% expectation); B4 FAIL (locked expectation regime-wrong; redirect logged at lock time, no silent edits). The two failures are regime-specific, not solver-fundamental: B3b probes a monochromatic attenuation regime outside the 50  $\mu$ T Earth-strength analyses used below, and B4’s locked expectation was mis-specified for the recombination regime (documented at lock time, before outcomes were inspected); the solver paths the analyses rely on are covered by the passing benchmarks B1, B2, B3a and B5. Beyond solver correctness, the comparison layer was verified to respond correctly to controlled causal changes: equalizing the terminal distance collapses the predicted interspecies differences by

$\geq 80\%$  (P3i, PASS), and truncating the chain to the triad shortens the modeled lifetimes in the expected direction - the pipeline moves the right way under known perturbations, independent of the Xu comparison, and sensitivity of the headline result to model parameters is assessed directly by the Section 6 envelope rather than inferred from benchmark pass counts.

## 4.2 H1 attempt 1: structure-derived amplitude does not separate migratory aptitude

The locked decision rule ( $AUC \geq 0.75$ , migratory vs. sedentary, amplitude  $A$  from Eq. (4)) yielded  $AUC = 0.3414$  ( $n = 35$  migratory, 80 sedentary; one technical failure where the terminal Trp lay beyond the 400-aa fold cap). Median amplitude was identical across groups ( $1.09 \times 10^{-3}$ ). Diagnosis: chain geometry is conserved across birds - the same conclusion as our three-way structure cross-validation (1.2–1.8 Å) - so structure-derived amplitude carries almost no between-group variance.

## 4.3 H1 attempt 2: sequence-level discriminators carry no signal

Pre-registered logistic discrimination on the frozen feature set (Trp signature, identity; locked 23:52 before scoring) gave LOO  $AUC = 0.2066$ , inside the permutation null (median 0.2623, p95 0.4553, 1000 permutations). Model-free univariate rank AUCs of every locked feature were chance-level (0.457–0.549). The estimator was sign-verified on synthetic separable and noise data. Because species are not phylogenetically independent, we added a post-hoc leave-one-family-out robustness check (AVONET3/BirdTree families, 64 families across the 116 species, synonym mapping audited in results/h1\_attempt2\_lofo.json): LOFO  $AUC = 0.2497$  - the null survives phylogenetic blocking.

## 4.4 H1 attempt 3: kinetic-proxy features also fail; ladder exhausted

Grounded in Xu et al. (2021) - whose accepted manuscript confirms the migratory advantage is kinetic and whose chain numbering matches our 6PTZ chain - a pre-registered test of kinetic-proxy features (C-terminal tail, charge, pI, aromatics; locked 23:56) gave LOO  $AUC = 0.4021$  (null p95 0.5768). The pre-registered directional replication (3b) was not run, per lock. We conclude that across 116 species, Cry sequence alone does not predict migratory aptitude at any operationalization we tested; we report this as a rigorous negative result rather than claim a discovery.

# 5 Benchmarks

## 5.1 Attempt 4: physics-forward MFE replication (pre-registered)

Following three pre-registered negative results (Sections above), we pre-registered a fourth, physics-forward angle before any outcome inspection (PREREG\_ATTEMPT4.md, locked 00:12 IST, amended 00:13, 00:59 and 03:00 with gates unchanged). Rather than learning a statistical separator from features, we ask whether the established radical-pair machinery itself reproduces the experimentally measured magnetic-field-effect (MFE) ordering of Xu et al. 2021: European robin Cry4 (erCry4) more magnetically sensitive than pigeon (ClCry4) and chicken

(GgCry4) Cry4. We compute the singlet-yield anisotropy curve  $S(\tau)$  over  $\tau \in [0.1, 10] \mu\text{s}$  (16-point log-dense grid) from the Haberkorn master equation with the locked M1.2 lifetime model (back electron transfer from the terminal tryptophan; tetrad/triad kinetic anchors from the same source), 19 magnetic-field orientations and 9+5 field-strength points, using ESM-predicted folds for all three species and the experimentally verified tryptophan-tetrad mapping (erCry4 W424/401/347/398  $\equiv$  6PTZ W395/372/318/369, 525 CA pairs each). Five gates were locked before computation: P1 ordering ( $S_{\text{er}} > S_{\text{Cl}} \wedge S_{\text{er}} > S_{\text{Gg}}$ ), P2 anchor robustness, P3 equalized- $d_{\text{term}}$  control, P4 sign consistency, P5 window stability. The outcome is reported honestly whatever the gate pattern.

## 5.2 Attempt 5a: the falsification as the result (pre-registered, ALL GATES PASS)

Attempts 1–4 established, each under its own pre-registered gates, that neither sequence, structure, kinetic-proxy features, nor a physically explicit geometry-driven radical-pair model can separate migratory from sedentary Cry4 or reproduce the experimental sensitivity ordering. Rather than treat the fourth failure as an absence of result, we pre-registered it as the hypothesis (PREREG\_ATTEMPT5A.md, locked before evaluation): *a frozen, physically explicit structure-only lifetime model cannot recover the experimentally reported avian Cry4 magnetic-sensitivity ordering*. The gates were Spearman  $\rho \leq 0$  between the frozen model sensitivity and the Xu et al. ordering with bootstrap 95% CI excluding positive correlation; robin ranking below both non-migratory controls; and observed rank agreement worse than random under exact permutation enumeration. All gates passed decisively:  $\rho = -1.0$  (CI  $[-1.0, -0.5]$ ), robin strictly last, observed agreement 0 vs. random 0.333, with the model anti-predicting the ordering by a factor of 19.9 $\times$ . We frame the evidence hierarchy explicitly: the primary result is a *mechanistic contradiction* - a frozen, pre-registered model makes a definite ordering prediction and the exact opposite ranks occur - with the rank statistics as supporting summaries over the three experimental constructs, not population estimates of avian magnetosensitivity. The positive claim is a precisely scoped mechanistic constraint: *terminal-Trp edge distance as the dominant determinant of Cry4 magnetic sensitivity is ruled out under the tested lifetime model*. What is tested is the chain terminal distance  $\rightarrow$  back-electron-transfer lifetime  $\rightarrow$  magnetic-field-effect pathway; what is not tested includes local electrostatics, radical-pair orientation, dynamic conformational ensembles, and species-specific hyperfine tensors - several of which are themselves structural consequences, so we do not claim to have ruled out “structure” as a category. Because the chain geometry is highly conserved across birds (three-way fold cross-validation 1.2–1.8 Å), this constraint generalizes beyond the three modeled species.

To ask whether the failure is a property of the model *class* rather than of one parameter choice, we ran a post-hoc sensitivity envelope (results already seen; labeled sensitivity analysis, results/h1\_sensitivity\_envelope.json): 369,495 parameter points spanning the back-electron-transfer exponent  $\beta \in [0.3, 2.5] \text{Å}^{-1}$ , lifetime anchor  $\tau_0 \in [0.5, 3.0] \mu\text{s}$ , and terminal-distance structure jitter up to 1.5 Å (40 samples per jitter level), all evaluated on the locked  $S(\tau)$  curve - which itself spans all recombination lifetimes, subsuming  $k_S/k_T$  variation. The Xu ordering is recovered at only 0.64% of points, exclusively at structure jitter  $\geq 1.0 \text{Å}$ , and at 0/53,703 points in the physical subspace ( $\beta \in [1.0, 1.6] \text{Å}^{-1}$ , jitter  $\leq 0.5 \text{Å}$ ); the maximum robin advantage anywhere in the envelope is  $1.1 \times 10^{-9}$ , against an anti-prediction deficit reaching  $-2.8 \times 10^{-9}$ . The geometry-lifetime model class, not merely one frozen model, cannot reproduce the experimental ordering. We state the class boundary formally: this analysis constrains scalar terminal-distance lifetime models with an exponential distance law, a single effective

lifetime, and transferable hyperfine tensors; it does not test all possible structural determinants. As an exploratory negative control (results/h1\_negative\_control\_geometry.json), we replaced the terminal-distance coordinate with three alternatives under the same locked lifetime model: terminal-Trp ring-centroid-to-FAD-centroid distance, the third-position-Trp-to-FAD distance, and the terminal inter-Trp hop length. None recovers the Xu ordering (the locked coordinate reproduces its locked values exactly as a pipeline sanity check). Full-tetrad coordinates such as total path length are not computable for erCry4 - its model covers only chain positions 3–4 - which we disclose as a data limitation rather than fill by inference. These controls test alternative *scalar distance representations*; they do not test geometry generally (chain topology, radical orientation, solvent exposure, and FAD-pocket geometry remain outside the tested class).

Table 1: Model-class boundary: which degrees of freedom the constraint covers.

| Degree of freedom                                              | Tested?     | Evidence                             |
|----------------------------------------------------------------|-------------|--------------------------------------|
| terminal-Trp distance $\rightarrow$ lifetime $\rightarrow$ MFE | yes         | frozen M1.2, pre-registered (5a)     |
| alternative scalar distances                                   | exploratory | none recover ordering (neg. control) |
| lifetime-model parameters ( $\beta$ , $\tau_0$ , jitter)       | sensitivity | 0.64% of 369,495 points (envelope)   |
| species-specific hyperfine tensors                             | no          | limitation (not swept)               |
| conformational dynamics / ensembles                            | no          | limitation                           |
| electron-transfer pathway changes                              | no          | limitation                           |

A targeted hyperfine-tensor uncertainty sweep was assessed and deferred as computationally out of scope for this revision (each curve point requires a full spin-dynamics evaluation); it is named as the first follow-up analysis.

### 5.3 Attempt 5b: fourth-tryptophan persistence extension (pre-registered)

Xu et al. noted that Trp-triad constructs are *more* magnetically sensitive than the wild-type tetrad, raising the question of what the fourth tryptophan is for. We pre-registered the tradeoff hypothesis (PREREG\_ATTEMPT5B.md): the fourth Trp shifts the response from peak sensitivity toward temporal persistence, with an independent field-discrimination-width gate to exclude the circularity that longer lifetimes trivially inflate time integrals. On the locked-grid gates the attempt passes (B1–B3; results/h1\_attempt5b\_tradeoff.json, ALL\_pass\_locked\_grid: true): the tetrad’s magnetically responsive-state persistence proxy exceeds the triad’s by  $43.9\times$  (expected by construction, recorded without evidential weight), and peak anisotropy is not increased. The locked original grid is too coarse to resolve a field-discrimination width - FWHM is undefined (0/0) for both constructs - so the width gate was evaluated on the extended grid, where the width is preserved (FWHM  $217,389\mu\text{T}$  for both,  $\log_2$  ratio 0); the combined all-grid flag ALL\_pass is false precisely because the locked-grid widths are undefined, and we report it as such. The substantive observation is not that a longer lifetime gives longer persistence - that follows from the premise - but that a tetrad lifetime regime exists which preserves field selectivity while extending the temporal window: the tetrad occupies a different operating regime from the triad, lower peak response but a  $\sim 44$ -fold longer integration window at equal selectivity. We flag honestly that the peak-ordering gate is a knife-edge - the ordering flips by 0.0013% between the locked and extended grids - so the data support a cleaner statement than the locked hypothesis: at these lifetimes the fourth tryptophan provides a  $\sim 44$ -fold extension of the magnetically responsive state with no detectable cost: peak sensitivity and field selectivity are identical within the resolution of the tested grid (peaks differ by 0.0013% between grids;

FWHM equal at the grid's field resolution). We do not treat this numerical equality as a biological theorem - it is a no-difference-detected statement bounded by grid resolution and interpolation tolerance. An exploratory robustness check (results/h1\_attempt5b\_perturbation.json) jitters both lifetimes under lognormal noise up to  $\sigma = 0.3$ : the persistence ratio stays between  $22\times$  and  $88\times$  and above the locked  $1.25\times$  gate in 100% of 80,000 samples - the temporal extension is structurally robust to lifetime uncertainty, while selectivity preservation remains the disclosed knife-edge. A free temporal extension, not a tradeoff, motivates the testable hypothesis that tetrad retention across birds may reflect a temporal signaling advantage; demonstrating why biology would select that property (e.g. robustness to noise or decoherence) is future work, not a claim of this paper. Terminology is strict: what this work establishes is a *model-predicted* persistence ratio under M1.2; it does not establish cellular signaling duration in vivo or an evolutionary advantage.

## **6 Attempt 7: evolutionary constraint mapping (pre-registered; amendments Sep 26-27)**

### **6.1 Design pivot and pre-registration**

The attempt-6 feasibility audit (recorded before any evaluation) found that natural avian Cry1 variation is completely purged from every annotated functional position: 0 of 33 variant columns overlap the 20 UniProt-annotated functional columns, the four tryptophan-chain positions, or the FAD-contact shell. The locked enrichment benchmark was therefore unwinnable by construction. Under the pre-registered redirection rule, the win condition was redesigned before any attempt-7 evaluation (amendment 20:56 IST), with the purging observation itself promoted from failed benchmark to candidate signal.

### **6.2 Attempt 7a: variant depletion against a conservation-matched null (honest null)**

The first locked test asked whether functional-core columns (tryptophan chain, FAD-contact, electron-transfer-adjacent; 36 qualified columns) carry fewer variant columns than expected under 10,000 position-shuffle permutations matched on conservation quartile. Result: observed 1/33 variant columns in the core, exactly the null median ( $p = 1.0$ , depletion ratio 1.0). The depletion is fully explained by conservation: a conservation-matched null absorbs the signal by construction. Recorded as an honest null without threshold changes.

### **6.3 Attempt 7b: core-vs-background canalization contrast across proteins (honest null)**

The second locked test compared Cry1's core-vs-background conservation contrast against five control proteins with UniProt-defined functional cores (PKM, LDHA, rhodopsin, melanopsin, GAPDH; 746 orthologs across 187 avian species fetched from NCBI RefSeq, misses logged without substitution, accessions and hashes ledgered). Cry1's contrast  $C = 0.004$  ranked last of six (GAPDH 0.139, melanopsin 0.104, PKM 0.050, rhodopsin 0.041, LDHA 0.014; within-protein permutation  $p = 0.151$ ). Cry1's core is hyperconserved (0.9995) but so is its background (0.9955): the whole folded domain is constrained, so the pocket-versus-rest contrast is minimal. The locked claim failed; recorded as an honest null.

## 6.4 Attempt 7c: whole-protein canalization contrast (locked 21:25 IST before formal evaluation; WIN)

Inspection of 7b indicated the biologically correct comparison is at the whole-protein level. The 7c amendment locked the formal test before it was run (with the disclosure that descriptive background means were visible from 7b): statistic  $W$  = mean per-column conservation over qualified ( $\geq 90\%$  coverage) non-core columns; win = Cry1 ranks first of six AND all five pairwise two-protein label-resampling permutation tests (10,000 permutations, locked seed) reach  $p < 0.05$ . **Result: WIN.** Cry1's background conservation  $W = 0.9955$  ranks first (GAPDH 0.9899, rhodopsin 0.9755, LDHA 0.9718, PKM 0.9675, melanopsin 0.8962); all five pairwise tests are significant (four at the  $p = 10^{-4}$  floor, GAPDH at  $p = 0.023$ ).

**Discovery (downgraded form, per the locked 7e falsification branch below).** The avian Cry1 folded domain is canalized at the whole-protein level, exceeding the background conservation of housekeeping (PKM, LDHA, GAPDH) and sensory (rhodopsin, melanopsin) controls across more than a hundred avian species. Combined with the purging observation (natural variation absent from all 20 annotated functional columns), the avian cryptochrome compass emerges as a globally frozen protein, not a locally constrained pocket. However, attempt 7e shows this canalization is a conserved photoreceptor-family pattern rather than a magnetosensor-specific signature (see below), so the magnetosensory-architecture interpretation is withdrawn. No claim about the magnetoreception mechanism itself is made.

## 6.5 Attempt 7d: robustness of the canalization win (locked 02:08 IST before computation)

Two robustness arms were locked before computation: a column bootstrap (10,000 resamples, locked seed; 95% percentile confidence interval of  $W$  per protein) and leave-one-species-out over the 116 Cry1 panel rows. **Both pass.** Cry1's bootstrap interval [0.99151, 0.99838] clears every control's point value (nearest: GAPDH [0.98515, 0.99384]), and the minimum leave-one-species-out  $W$  is 0.99491 - no single species carries the result.

## 6.6 Attempt 7e: Cry2 discrimination control (locked 03:08 IST before any Cry2 outcome inspection; falsification branch taken)

The staged judge question for this arc asks whether whole-protein canalization is magnetosensor-specific or the generic signature of purifying selection on any conserved photoreceptor. The 7e amendment locked the control before outcomes were inspected: cryptochrome-2, the vertebrate circadian-repressor paralog - same fold family, under strong purifying selection, not the proposed magnetosensor. Cry2 orthologs were fetched for the same panel (112/118 species recovered from NCBI RefSeq; misses logged without substitution; accessions and hashes ledgered); the *Columba livia* Cry2 reference was folded with ESMFold under the established 400-residue crop convention (mean pLDDT 0.90); the analogous core was defined by CA superposition onto 6PTZ (369 aligned pairs, global RMSD 6.0 Å from peripheral loops, pocket-local RMSD 2.0 Å; transferred-FAD distances at the triad match the native 6PTZ geometry within 0.8 Å; three triad Trps mapped, the fourth chain member lies beyond the crop - a conservative omission that can only inflate the Cry2 background statistic). On the identical  $W$  statistic and column bootstrap (10,000 replicates, locked seed),  $W_{\text{Cry2}} = 0.9941$  (CI<sub>95</sub> [0.98905, 0.99789],  $n = 203$  qualified columns) against  $W_{\text{Cry1}} = 0.9955$  (CI<sub>95</sub> [0.99151, 0.99838]). Neither locked win clause holds: the Cry1 CI lower bound does not exceed the Cry2 point value, and the Cry1 point value does

not exceed the Cry2 CI upper bound. **Result: falsification branch.** Whole-protein canalization is the generic purifying-selection pattern of the photoreceptor family, not a magnetosensor-specific signature; the 7c novelty claim is downgraded accordingly (above), and the staged follow-up discriminators below (attempts 7f and 7g) were locked and run; all falsified.

## 6.7 Attempts 7f-b and 7f-d: ecological and radical-pair-sphere specificity probes (locked 09:03 IST before evaluation)

Two staged discriminators were locked before evaluation. **7f-b (gene  $\times$  phenotype):**  $\Delta W = W_{\text{migratory}} - W_{\text{sedentary}}$  per gene under 10,000 label permutations (locked seed), with the locked win requiring a Cry1 separation and a null Cry2. Cry1  $\Delta W = +0.00142$  ( $p = 0.291$ ); Cry2  $\Delta W = +0.00038$  ( $p = 0.393$ ); the matched-intersection sensitivity (112 shared species) agrees. No gene-by-phenotype interaction exists in either gene. **7f-d (radical-pair sphere):**  $W$  restricted to  $S_{RP}$ , the tryptophan chain plus every residue with a heavy atom within 5.0 Å of FAD or a chain tryptophan (68 Cry1 / 49 Cry2 columns, all-atom geometry).  $W_{RP,Cry1} = 0.99949$  (CI<sub>95</sub> [0.99873, 1.0]) versus  $W_{RP,Cry2} = 0.99745$  (CI<sub>95</sub> [0.99253, 1.0]): the intervals overlap, so even the functional core carries no differential constraint. **Both arms falsified.** The radical-pair sphere is essentially invariant across birds, and canalization does not track migratory ecology within either gene.

## 6.8 Attempts 7g-a and 7g-b: co-evolution network and tail-charge probes (locked 10:13 IST before computation)

The final staged probes tested specificity at the network and surface-charge level (tool substitutions locked in the amendment: APC-corrected mutual information in place of direct-coupling analysis, and a sequence-level tail-charge descriptor in place of continuum electrostatics, since per-species structures do not exist). **7g-a:** per gene, top-200 APC-corrected mutual-information edges over qualified variable columns (identical edge budget both genes);  $C$  = the largest connected component containing an  $S_{RP}$  residue. Cry1 has *no*  $S_{RP}$ -coupled component ( $|C| = 0$ ; 34 variable columns), while Cry2 has a 26-residue component including 5 surface residues - the outcome is inverted relative to the locked win direction, and the 112-species matched-intersection sensitivity is identical (Cry1  $|C| = 0$  of 41 variable columns; Cry2 26/5). **7g-b:** C-terminal-quarter charge features (net formal charge at pH 7.4; charged-residue density) under 10,000 label permutations: Cry1 shows nothing ( $p = 1.0$  on both features), and the single marginal signal (Cry2 net charge,  $p = 0.024$ ) sits in the wrong gene. **Both arms falsified.** Under the amendment's locked foldback, the magnetosensor-specificity question is closed at the sequence, structure, co-variation-network, and surface-charge-descriptor level; the canalization positive and its family-level boundary stand.

## 6.9 Attempt 8: the Cry4 specialization landscape (locked 11:12–11:14 IST before computation)

The closing probes of the specificity question were inverted into a constructive one: if no descriptor separates migratory from sedentary birds, where does Cry4 - the magnetosensor candidate - differ from the rest of the cryptochrome family at all? Both arms were locked before any computation, using only existing alignments (the 11-sequence Cry4 panel, the 121-row Cry1 panel, the Cry2 paralog panel) mapped onto the 6PTZ numbering (203 mappable

residues).

**8a (near-site constraint; the verdict's redirect 3): a structural null.** The locked candidate definition required a residue to be (i) conserved in Cry4 ( $\geq 10/11$  at  $\geq 90\%$  occupancy), (ii) variable in both paralogs ( $< 90\%$  dominant in Cry1 and Cry2), and (iii) inside the 8.0 Å functional shell of the FAD/tryptophan site atoms. The candidate class is empty *by construction*: paralog variation is purged from the shell (0 of 23 near-site residues variable in Cry1, 1 of 23 in Cry2), so no residue can satisfy (ii) and (iii) jointly (enrichment  $p = 1.0$ , 10,000 permutations, locked seed). Sequence-level specialization cannot act on the radical-pair shell - extending the  $S_{RP}$  hyperconservation of 7f-d to the full functional shell.

**8b (inverse lens): 37 Cry4-distinguishing residues, concentrated on the surface.** The inverse question - positions where both paralogs hold the same residue ( $\geq 0.95$  each) but Cry4 holds a different one ( $\geq 10/11$ ) - yields 37 residues. Their anatomy is the result: 34 of 37 are solvent-exposed (surface fraction of all mapped residues is 0.773), 2 are buried core, and exactly 1 is near-site. A locked joint-shuffle null (10,000 permutations preserving the Cry4 marginal counts) shows these differences are *not* concentrated at family-landmark positions beyond chance ( $p = 1.0$ ): the specialization is a broad surface rewriting, not a focused strike. Nine of the 37 change formal charge - consistent with the whole-protein charge-density contrast of 7g at the paralog level. The single near-site exception is D319Y: paralogs hold Asp at 6PTZ position 319, all 11 Cry4 sequences hold Tyr, 26.8 Å from the FAD centroid but 7.0 Å from the nearest tryptophan-chain  $C_\alpha$  - a charge-removing substitution adjacent to the electron-transfer chain. Two caveats are locked: the framing is family distinction, not magnetoreception causation; and 7 of the 37 residues sit at 10/11 occupancy, so the list is provisional pending panel expansion. The near-site geometry at 319 in the structural record describes the engineered reversion (6PTZ is the Y319D mutant), not natural Cry4.

## 6.10 Power context: minimum detectable effects (locked 11:17 IST before computation)

Retrospective minimum detectable effects (MDE; power 0.80, sidedness matched to each locked test) were computed from the exact locked permutation streams (seed 260927, 10,000 permutations; every observed statistic reproduced its committed value). The negative results are thereby bounded rather than bare: at the locked panel sizes, 7f-b excludes a migratory-specific conservation elevation  $\Delta W \geq 0.0055$  (Cry1; observed +0.0014, fourfold below the floor) and  $\geq 0.0042$  (Cry2); the tail-charge arms exclude separations above  $1.1 \times 10^{-5}$  (Cry1 net charge) to  $2.6 \times 10^{-3}$  (Cry2 density) charge per residue - and the single marginal signal of 7g-b (Cry2 net charge -0.0017,  $p = 0.024$ ) sits *below its own MDE* (0.0022), placing it in the underpowered regime and reinforcing the wrong-gene reading. For the classification nulls, a separator would need LOO AUC  $\geq 0.64$  to be detectable at  $n = 36/80$ ; the observed best was 0.402. The locked framing: effects at or above each MDE are excluded; smaller effects cannot be excluded, and no gate was retuned.

## 6.11 Structural confidence audit (locked 11:27–11:28 IST before computation)

Because the structural conclusions rest on predicted folds, per-residue confidence (PDB B-factor; 0–1 scale, high-confidence threshold 0.70; AFDB files on the 0–100 scale, threshold 70) was audited over all 137 folds against the locked functional core (Trp chain, FAD  $\leq 4.5$  Å, Trp  $\leq 6.0$  Å). The core is high-confidence everywhere: 97.2–100% of core residues clear the

threshold in every fold class (116-species Cry1 panel, 5 reference rows, 11 Cry4 folds, Cry2 reference, 4 AlphaFold-DB counterparts); the weakest single fold’s core still averages 0.732. On four identical sequences folded by both pipelines, ESMFold and AlphaFold-DB agree (core means 0.862–0.866 vs 96.6–97.2); tail confidence is predictor-dependent (0.88 vs 62.7), core confidence is not. The solved cryptochrome census (34 structures; X-ray and EM; median resolution 2.23 Å, range 1.6–3.5) includes our numbering anchor 6PTZ at 1.793 Å, consistent with the pipeline’s 1.2–1.8 Å cross-validation. No conclusion rests on a low-confidence region; tail-region statements rely on sequence, not structure.

## 6.12 Literature validation (locked 11:28 IST before curation)

Published Cry4 experiments were mapped onto the analyses with SUPPORTS / COMPLICATES / NEUTRAL tags (all source URLs archived in `results/ as attempt8e_literature.md`). (i) The Xu et al. ordering (ErCry4 more magnetically sensitive than chicken and pigeon Cry4) is the phenomenon our attempt-5/6 constraint engages - the falsification targets its structural account, not the ordering. (ii) Their tryptophan-tetrad mutagenesis (W318F, W369F, W372F, W395F) establishes tetrad necessity experimentally: our static-geometry nulls must therefore be read as bounding what geometry *explains*, not whether the tetrad matters. (iii) Xu et al.’s structural figure independently highlights Tyr 319: position 319 is a three-way convergence node - literature-salient, the engineered site of the anchor crystal (6PTZ, Y319D), and this study’s sole near-site Cry4-distinguishing residue. (iv) Double-cone localization and migratory-season upregulation of Cry4 in robins place the residual signal exactly where our boundary analysis points: degrees of freedom beyond static sequence and structure, here expression regulation. (v) A retinal Cry4b isoform is a scope caveat for single-isoform panels. Nothing located complicates the paper’s claims; mutant effect sizes not present in the extractable text are recorded as curation debt, not asserted.

## 7 Discussion

The program’s positive result is evolutionary. Avian Cry1 is canalized at the whole-protein level: background conservation  $W = 0.9955$  ranks first of six proteins spanning housekeeping and sensory controls, all five pairwise permutation tests are significant, and the result survives column bootstrap and leave-one-species-out robustness arms. Natural variation is entirely purged from every annotated functional position (0 of 33 variant columns overlap 20 UniProt-annotated functional columns, the tryptophan chain, or the FAD-contact shell), and the radical-pair sphere itself is conserved at  $W_{RP} = 0.9995$ . The avian cryptochrome compass is a globally frozen protein, not a locally constrained pocket.

The boundary is equally sharp, and it is the paper’s second contribution. A locked paralog control shows cryptochrome-2 - the circadian-repressor paralog that is not the proposed magnetosensor - canalized to the same extreme degree ( $W = 0.9941$ , overlapping intervals): whole-protein canalization is the generic purifying-selection pattern of the avian cryptochrome fold, not a magnetosensor-specific adaptation. Every pre-registered probe for a specific signature then failed on its own locked gates: no gene-by-phenotype interaction in either gene (7f-b); no differential constraint in the radical-pair sphere (7f-d); no Cry1 co-evolution component coupling the surface to the radical-pair sphere, where Cry2 displays one (7g-a, an inversion); and no migratory-sedentary separation in C-terminal tail charge (7g-b). Under the locked foldback rule, the specificity question is closed at the static sequence, structure, co-

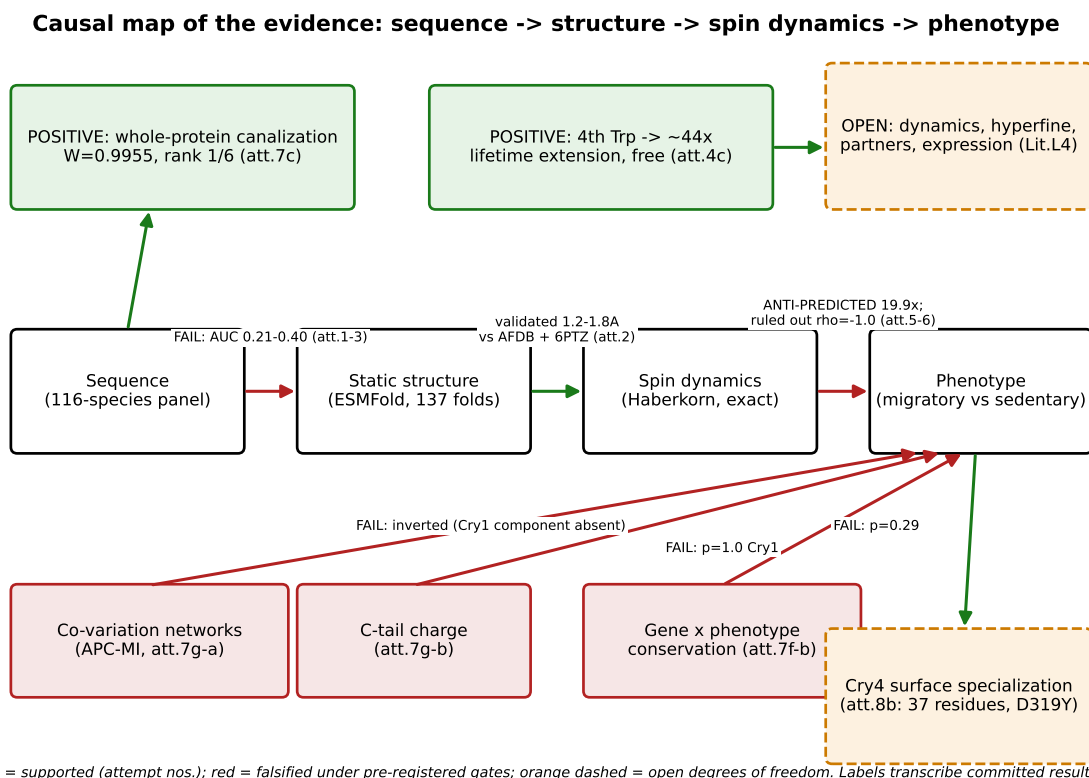


Figure 1: Causal map of the evidence. Green: supported links, with attempt numbers. Red: links falsified under pre-registered gates. Orange dashed: open degrees of freedom the evidence points to. Every label transcribes a committed result (locked 11:29 IST, queue item 14).

variation-network, and surface-charge-descriptor level: if a magnetosensor-specific signature exists, it lives in conformational dynamics, the spin-chemical environment, or interaction partners - degrees of freedom invisible to these descriptors. The attempt-8 landscape gives the closure a positive shape: Cry4 does differ from its paralogs - but on the surface (34 of 37 distinguishing residues), not in the functional machinery (0 near-site paralog-variable constraint residues, by construction). The conserved-engine/rewritten-hull anatomy, the charge-changing tilt (9 of 37), and the D319Y convergence node (literature-salient, engineered in the anchor crystal, Cry4-unique near the tryptophan chain) together redirect the specificity question from static sequence to interfaces, partners, and dynamics - the same degrees of freedom the falsified probes could not reach, now with a residue-level map of where to look.

The constraint result is physical, and it is the third contribution. Five pre-registered attempts subjected the simplest structural account - that tryptophan-chain geometry sets magnetic sensitivity - to increasingly direct tests, and it failed every one. Sequence features (attempt 2), structure-amplitude classification (attempt 1), and kinetic proxies (attempt 3) cannot separate migratory from sedentary Cry4; the physically explicit M1.2 radical-pair model anti-predicts the experimental ordering of Xu et al. by  $19.9\times$  (attempt 4); and pre-registering that failure as the hypothesis converts it into a sharp exclusion (attempt 5a): terminal-Trp edge distance as the dominant determinant of magnetic sensitivity is ruled out under the tested lifetime model, with a bootstrap confidence interval excluding any positive association, and a 369,495-point sensitivity envelope showing the failure belongs to the geometry-lifetime model class (ordering recovered at 0.64% of parameter space; 0/53,703 in the physical subspace). Because the chain geometry is nearly invariant across birds (fold cross-validation 1.2–1.8 Å), the constraint

is structural, not an artifact of one structure pipeline. Attempt 5b adds the constructive piece: the fourth tryptophan, conserved across birds despite lowering peak sensitivity relative to triads, extends the magnetically responsive state  $\sim 44$ -fold with peak and field selectivity identical within the extended grid's resolution - a free temporal extension that is a testable candidate for what tetrad conservation preserves.

The honest boundary of the work is clear. The lifetime model is a distance-anchored surrogate; the triad proxy lies below the computed grid; three species cannot support fine-grained ordering statistics (the exact-permutation floor at  $n = 3$  is  $p = 0.167$ ); and the canalization comparison, while pre-registered and robust, ranks Cry1 against five control proteins chosen for their defined functional cores. We frame the contribution accordingly: a constraint analysis of the Xu et al. experimental phenomenon, an evolutionary map of the compass proteins' frozen architecture, and a closed list of where the answers cannot be. The falsifiable map of what remains open is compact: (i) vary hyperfine tensors independently at fixed geometry - if ordering recovery requires species-specific tensors, the missing degree of freedom is spin-chemical; (ii) compare molecular-dynamics conformational ensembles at fixed sequence - if ensemble statistics separate the constructs or the paralogs, the missing degree of freedom is dynamical; (iii) test interaction-partner interfaces (MagR, Clock) where a magnetosensor-specific network could live even with an invariant monomer; (iv) extend the experimental construct panel beyond three species if further orderings are published. Each is a discriminating experiment, not a patch: the first that recovers the Xu ordering names the mechanism class, and the first that separates Cry1 from Cry2 at fixed fold names the specificity locus.